Euclid's Elements — Book I
Proposition 25

Formal Reconstruction — Technical Documentation

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A greater third side gives a greater included angle

1 Introduction

Euclid's Proposition I.25 is the converse comparison theorem to Proposition I.24.

Suppose two triangles have two corresponding sides congruent:

$$AB \cong DE, \qquad AC \cong DF.$$

If the third side of the first triangle is greater than the third side of the second,

$$EF < BC,$$

then the included angle of the first triangle is greater:

$$\angle EDF < \angle BAC.$$

The formal reconstruction is strikingly short. Unlike Proposition I.24, no auxiliary geometric construction is needed inside the proposition itself. The proof is an elimination argument based on the previously established trichotomy for strict angle comparison.

The difficult geometric work needed for I.25 has already been absorbed into the Hilbert interface, in particular into `angle_trichotomy`. Once that theorem is available, Proposition I.25 becomes almost purely logical.

2 The Classical Configuration

Let $\triangle ABC$ and $\triangle DEF$ satisfy

$$AB \cong DE, \qquad AC \cong DF,$$

and suppose

$$EF < BC.$$

The conclusion is

$$\angle EDF < \angle BAC.$$

No new point is constructed by the proof body of Proposition I.25. The same two triangles are compared throughout the trichotomy argument. The strict segment-order hypothesis, however, contains its own hidden geometric witness; this is separated below from proposition-specific construction.

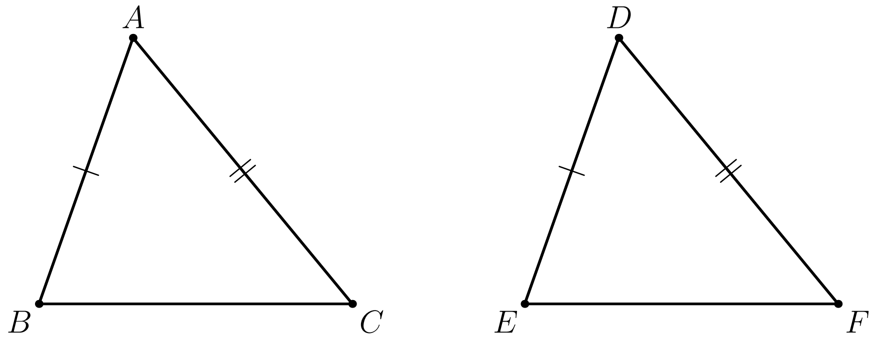

Figure 1: The classical configuration for Proposition I.25. The two pairs of corresponding sides $AB \cong DE$ and $AC \cong DF$ are marked. The hypothesis $EF < BC$ is not encoded by the apparent drawn lengths; strict segment order is a synthetic relation and is expanded explicitly in Figure 2.

3 Diagrammatic Audit

The preserved Book1R PDF contains one figure for I.25. It correctly records the two triangles and the two side-congruence classes, but the strict hypothesis $EF < BC$ is visible only through the accidental scale of the drawing. Such scale is not part of the synthetic proof language.

In the current formalization, `HilbertSegmentLess Geo E F B C` means that there exists a point H with

$$B - H - C, \qquad EF \cong BH.$$

Thus the formal input contains a genuine hidden witness even though the proof of `euclid_proposition_25` does not construct a new point. Figure 2 records this witness explicitly.

No additional figure is introduced for the three branches of `angle_trichotomy`. Those branches change only the logical relation between the same two included angles; they introduce no new point, line, betweenness relation, side relation, or auxiliary construction. Likewise, the internal geometry of Proposition I.24 is not repeated here: it belongs to the diagrammatic audit of I.24.

4 Classical Proof

Euclid excludes the two alternatives incompatible with the required conclusion.

First suppose

$$\angle BAC \cong \angle EDF.$$

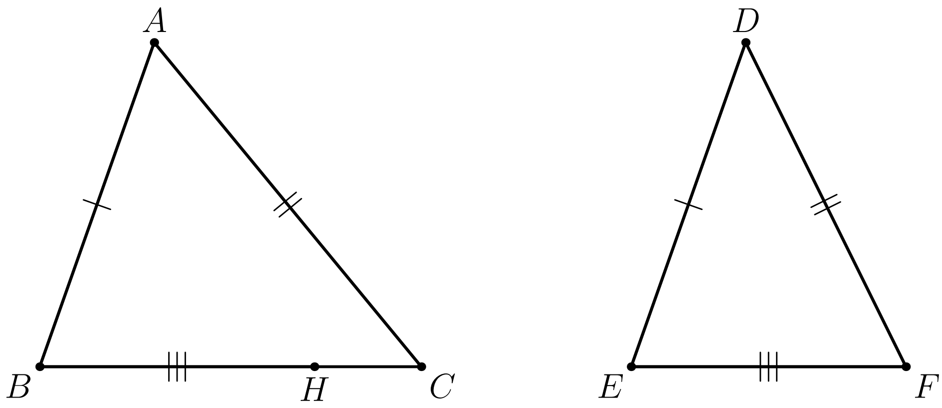

Figure 2: The exact geometric witness hidden in the strict segment-order hypothesis. The point H satisfies $B - H - C$ and $BH \cong EF$. Together with the original side congruences this is a synthetic realization of $EF < BC$; no numerical length comparison is used.

Then Proposition I.4 (SAS), applied to the two pairs of congruent sides, gives

$$BC \cong EF,$$

contradicting $EF < BC$.

Next suppose

$$\angle BAC < \angle EDF.$$

Apply Proposition I.24 with the two triangles interchanged. Since the included angle of $\triangle DEF$ is greater, I.24 gives

$$BC < EF,$$

again contradicting $EF < BC$.

Therefore the only remaining possibility is

$$\angle EDF < \angle BAC.$$

5 Angle Trichotomy

The decisive interface result is `angle_trichotomy`. For two nondegenerate angles it provides exactly the three alternatives

$$\angle BAC \cong \angle EDF, \qquad \angle BAC < \angle EDF, \qquad \angle EDF < \angle BAC.$$

This theorem was established in the Hilbert layer immediately before the formalization of I.25. Its proof uses Hilbert angle construction, same-side information, the order of rays in a half-plane, and the principle that two collinear points lying on the same side of a line through the vertex determine the same ray.

Thus angle trichotomy is not an additional axiom introduced for I.25. It is a derived structural theorem about the already defined relation `HilbertAngleLess`.

6 Proof Architecture of the Reconstruction

The formal proof is a three-branch elimination argument. Its complete architecture is short enough to display directly in ASCII:

```
angle_trichotomy
    |
    +-- angles congruent
    |      -> SAS
    |      -> BC ~= EF
    |      -> contradiction with EF < BC
    |
    +-- BAC < EDF
    |      -> Proposition I.24
    |      -> BC < EF
    |      -> contradiction with EF < BC
    |
    +-- EDF < BAC
           -> goal
```

This is not merely a summary of the Lean tactics. It is the actual mathematical dependency structure of the reconstructed proof.

The first branch is eliminated by triangle congruence. The second is eliminated by the direct hinge theorem I.24. The third branch is already the desired statement.

Consequently, after angle trichotomy has been established, the proof body of I.25 contains no proposition-specific construction and no additional geometric case analysis. The only hidden point relevant to the theorem statement is the witness already packaged inside the input relation $EF < BC$.

7 Hilbert Reconstruction

7.1 The Strict-Segment Input Witness

The hypothesis

`HilbertSegmentLess Geo E F B C`

is an existential synthetic statement. Unfolded, it supplies a point H such that

$$B - H - C, \qquad EF \cong BH.$$

This witness is part of the input data rather than a construction made by `euclid_proposition_25`. The public proof never needs to choose it explicitly because the generic segment-order lemmas `hilbert_segmentLess_not_congruent` and `hilbert_segmentLess_asymm` consume the strict-order relation directly.

7.2 The Congruent-Angle Branch

Assume

$$\angle BAC \cong \angle EDF.$$

Together with

$$AB \cong DE, \qquad AC \cong DF,$$

Hilbert SAS yields

$$BC \cong EF.$$

The interface theorem `hilbert_segmentLess_not_congruent` states that a strictly shorter segment cannot be congruent to the longer one. Hence this branch is impossible.

7.3 The Wrong Strict-Inequality Branch

Assume

$$\angle BAC < \angle EDF.$$

Interchange the roles of the triangles and reverse the side congruences:

$$DE \cong AB, \qquad DF \cong AC.$$

Then Proposition I.24 gives

$$BC < EF.$$

But the original hypothesis is

$$EF < BC.$$

The asymmetry theorem `hilbert_segmentLess_asymm` excludes these simultaneous strict inequalities.

7.4 The Desired Branch

The remaining case is exactly

$$\angle EDF < \angle BAC,$$

so no further argument is needed.

8 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_25
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E F : Geo.Point)
    (hABC : not (PrimCollinear Geo A B C))
    (hDEF : not (PrimCollinear Geo D E F))
    (hAB_DE : Geo.Congruent A B D E)
    (hAC_DF : Geo.Congruent A C D F)
    (hEF_BC : HilbertSegmentLess Geo E F B C) :
    HilbertAngleLess Geo E D F B A C
```

The proof begins by invoking `angle_trichotomy` on the two included angles.

In the equality branch, `SAS` produces a `TriangleCongruenceResult`; its field `sideBC` gives $BC \cong EF$. After reversing the congruence, the contradiction is discharged by `hilbert_segmentLess_not_congruent`.

In the second branch, Proposition I.24 is called with the two triangles interchanged. This gives $BC < EF$, contradicting the hypothesis through `hilbert_segmentLess_asymm`.

The third branch is returned directly.

9 Relationship with Earlier Propositions

9.1 Proposition I.4

Proposition I.4 supplies the SAS criterion used to eliminate the equality branch:

$$\angle BAC \cong \angle EDF, \quad AB \cong DE, \quad AC \cong DF \quad \Longrightarrow \quad BC \cong EF.$$

9.2 Proposition I.24

Proposition I.24 is the central direct dependency of I.25. I.24 states that, under two pairs of congruent sides, a greater included angle implies a greater third side. I.25 reverses the comparison: a greater third side implies a greater included angle.

The formal proof of I.25 uses I.24 exactly once, to eliminate the wrong direction of angle inequality.

Together they give, under

$$AB \cong DE, \qquad AC \cong DF,$$

the order correspondence

$$\angle EDF < \angle BAC \quad \Longleftrightarrow \quad EF < BC.$$

10 Hilbert and Book Zero Components Used

10.1 `angle_trichotomy`

This is the principal structural result exposed by I.25. It converts the comparison of two nondegenerate angles into an exhaustive three-way split: congruent, smaller in one direction, or smaller in the other direction.

10.2 SAS

The existing SAS theorem converts equality of the included angles and the two side congruences into congruence of the third sides.

10.3 Strict order versus congruence

This theorem expresses incompatibility between strict segment order and segment congruence. It eliminates the first trichotomy branch.

10.4 Asymmetry of strict segment order

This theorem expresses asymmetry of strict segment comparison:

$$AB < CD \quad \Longrightarrow \quad \neg(CD < AB).$$

It eliminates the second branch after I.24 produces the opposite inequality.

11 Formalization Notes

Proposition I.25 provides a useful contrast with I.24.

I.24 required substantial geometric work. Its formal reconstruction had to expose point-order cases hidden by Euclid's diagram and develop the corresponding hinge argument in each configuration.

I.25 does not repeat any of that proposition-specific geometry. Its strict segment-order hypothesis already packages a betweenness witness, but the proof itself does not construct or unpack that witness. Instead, the reconstruction first strengthens the reusable Hilbert interface by proving angle trichotomy. Once this has been done, I.25 is a direct logical consequence of

$$\text{angle trichotomy}, \qquad \text{SAS}, \qquad \text{I.24}.$$

This separation is important for the architecture of the project. A long foundational or order-theoretic argument should not be buried inside every later Euclidean proposition that needs it. Once identified, it becomes part of the intermediate synthetic language and later proofs become correspondingly transparent.

I.25 is therefore a particularly clean example of the role of the Hilbert interface: difficult geometric infrastructure is proved once, after which a classical theorem can recover its natural short proof.

12 Dependency Summary

The underlying dependency graph is

```
Hilbert incidence + order + congruence
        |
        v
same-side and ray-order theory
        |
        v
angle_trichotomy
        |
        +-------------------+
        |                   |
```

```
 v                   v
SAS            Proposition I.24
 |                   |
 +---------+---------+
           |
           v
euclid_proposition_25
```

At the level of the final theorem:

```
angle_trichotomy
    |
    +-- BAC ~= EDF
    |     -> SAS
    |     -> BC ~= EF
    |     -> contradiction with EF < BC
    |
    +-- BAC < EDF
    |     -> I.24
    |     -> BC < EF
    |     -> contradiction with EF < BC
    |
    +-- EDF < BAC
          -> exact
```

No coordinate system, numerical segment length, or numerical angle measure is introduced. The current production file contains no local `axiom` declaration and no `sorry`; this documentation audit does not claim a fresh transitive axiom audit because neither `lake build` nor `#print axioms` was run here.

13 Conclusion

Proposition I.25 is the converse comparison theorem to Proposition I.24.

Its formal reconstruction is exceptionally concise, but the concision is not accidental. The theorem becomes short only after the underlying angle-order theory has been completed.

The proof begins with angle trichotomy. Congruence of the included angles is excluded by SAS, because it would force congruence of the third sides. The wrong strict inequality is excluded by Proposition I.24, because it would force the opposite strict comparison of the third sides. The remaining branch is precisely the desired conclusion.

Thus I.25 gives a particularly clear instance of the architecture

$$\text{foundational Hilbert theory} \longrightarrow \text{reusable synthetic interface} \longrightarrow \text{short Euclidean theorem.}$$